SHRAVANI PRANAY SAWANT

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SUMMARY

M.S. Computer Science (Cybersecurity) graduate with research experience in distributed reinforcement learning, AI security, governance, and reliable AI systems. Designed and implemented a logical scaling framework using Ray RLlib to simulate partial worker failures and evaluate model convergence, learning stability, latency, and training performance across MuJoCo and Atari benchmarks. Developed privacy-aware AI agents and security analytics tools involving policy enforcement, PII detection, risk assessment, and human-in-the-loop controls. Skilled in Python, AI/ML evaluation, distributed systems, privacy engineering, and cybersecurity, with interests in trustworthy AI, responsible AI development, and AI risk management.

EDUCATION

STEVENS INSTITUTE OF TECHNOLOGY **May 2026**
Master of Science, Computer Science (Cybersecurity) *Hoboken, NJ*
• **GPA:** 3.8/4.0

GOVT. INSTITUTE OF FORENSIC SCIENCE **Jun 2024**
Bachelor of Science, Forensic Science (Digital & Cyber Forensics) *Mumbai*
• **GPA:** 9.2/10.0

RESEARCH EXPERIENCE

INTELLISYS LAB. **Oct 2025 - May 2026**
Research Intern (Graduate Student) *Hoboken, NJ*
Advisor: Dr. Hao Wang · Focus: Distributed RL Systems, Fault Tolerance & AI Infrastructure Reliability
- Investigated fault tolerance in distributed reinforcement learning systems through controlled partial worker failure experiments to assess learning stability, system reliability, and performance.
- Designed and implemented a logical scaling framework using Ray RLlib on Linux to simulate partial worker failures, test distributed AI system behavior, and evaluate learning under dynamic resource conditions.
- Conducted experimental testing and validation across MuJoCo and Atari benchmarks, analyzing convergence, reward progression, latency, checkpoint overhead, and overall training performance.
- Analyzed actor contribution variability and its implications for learning-aware fault tolerance, reliability, scalability, and risk assessment in distributed AI systems.

PROJECTS

PersonaGuard: Privacy-Aware AI Agent PersonaGuard
- Developed a privacy-aware AI agent that applies user-defined privacy policies through a governance engine and trust-based decision framework.
- Implemented real-time PII detection and privacy scoring to support risk-based substitution, blocking, and human-in-the-loop approval workflows.
- Investigated how configurable privacy policies influence AI agent behavior, privacy preservation and user trust.
- GitHub: PersonaGuard

SK_Scout: Behavioral Analysis of GitHub Open-Source Ecosystem SK_Scout
- Analyzed suspicious activity patterns within open-source ecosystems using GitHub event streams, repository metadata, and user activity timelines to identify behavioral risks.
- Investigated behavioral indicators associated with phishing repositories, bot activity, and coordinated account behavior.
- GitHub: SK_Scout

TrapTrack: Detection of Privacy Leaks in Cookie Consent Workflows TrapTrack
- Developed a browser instrumentation framework to analyze data-sharing behavior before and after cookie consent interactions.
- Examined third-party tracking activity, network requests, cookies, and browser storage artifacts across live websites.
- Investigated privacy compliance challenges and unintended data disclosures in modern web ecosystems.
- GitHub: TrapTrack

AegisAI: LLM-Assisted Cybersecurity Detection and Threat Analysis Platform
- Explored the use of large language models for phishing detection, malicious file assessment, and URL analysis.
- Combined rule-based detection techniques with LLM assisted reasoning to support explainable threat triage.
- Investigated opportunities and limitations of AI assisted security analysis.

SERI: Cyber Risk Assessment Framework for Financial Systems SERI

- Designed a structured framework to identify, categorize, and prioritize cyber risks across financial systems by analyzing assets, threat vectors, vulnerabilities, and existing controls.
- Developed risk matrices and mitigation mappings aligned with security governance, compliance, and risk management objectives.
- GitHub: SERI

SKILLS

- **Programming & Scripting**: Python, C/C++, Java, SQL, Bash, JavaScript
- **AI, ML & Systems**: Reinforcement Learning, Distributed Systems, Distributed ML Infrastructure, AI Agents, Information Retrieval, Performance Profiling, Data Mining, AI System Testing & Debugging, AI Failure Mode Analysis, AI Explainability & Guardrails, Model Evaluation
- **AI Governance, Risk & Security**: AI Security, NIST, ISO, Risk Management, Governance Frameworks, Privacy Engineering, Threat Modeling, Vulnerability Assessment, Risk & GRC, Internal Controls, Testing & Validation, Digital Forensics
- **Tools & Frameworks**: Ray RLlib, PyTorch, Git, GitHub Actions, Playwright, MITMProxy, Wireshark, Burp Suite, Splunk, Nmap, Autopsy
- **Web & Application Development**: HTML, CSS, Node.js, Express, React, Typescript
- **Databases & Platforms**: MongoDB, PostgreSQL, SQLite, AWS (EC2, Lambda, S3), Linux (Ubuntu/Kali), macOS, Windows

PUBLICATIONS

- Gender Specificity Amid Toys.*International Journal of Innovation Research in Technology (IJIRT) Vol. 11, 2024* Publication Link
- Blockchain Interoperability Using Hash Time Locks.*Proceeding of Fifth International Conference on Microelectronics, Computing and Communication Systems: MCCS, 2021* Publication Link